

Problem 93. Let I be the incentre of $\triangle ABC$ and let a circle Γ through the points A and B intersect the lines

- AI at points A and P ,
- BI at points B and Q ,
- AC at points A and R , and
- BC at points B and S ,

with none of the points A, B, P, Q, R and S coinciding and the points R and S being interior points of the line segments AC and BC , respectively.

Prove that the lines PS , QR and CI are concurrent.

Problem 94. On a circle 2020 points are marked. Each of these points is labelled with an integer. Let each number be larger than the sum of the preceding two numbers in clockwise order.

Determine the maximal number of positive integers that can occur in such a configuration of 2020 integers.

Problem 95. Determine all digits d such that for each integer $k \geq 1$ there exists an integer $n \geq 1$ with the property that the decimal representation of n^9 ends with at least k digits d .

Problem 96. Consider a board consisting of $n \times n$ unit squares where $n \geq 2$. Two cells are called neighbours if they share a horizontal or vertical border. In the beginning, all cells together contain k tokens. Each cell may contain one or several tokens or none.

In each turn, choose one of the cells that contains at least one token for each of its neighbours and move one of those to each of its neighbours. The game ends if no such cell exists.

(a) Find the minimal k such that the game does not end for any starting configuration and choice of cells during the game.

(b) Find the maximal k such that the game ends for any starting configuration and choice of cells during the game.